Project: Pizza Sales Analysis

Project Overview

The Pizza Sales Analysis project is designed to analyse transactional sales data from a pizza store. The goal is to identify key business insights, trends, and KPIs that will help management make informed decisions related to sales, marketing, and operations.

Business Objectives

- Identify overall revenue, total pizzas sold, and total number of orders.
- Determine sales distribution by pizza category, size, and type.
- Analyse time-based trends in sales (daily, monthly, and yearly).
- Highlight best-selling and least-selling pizzas by revenue and quantity.
- Understand customer purchasing behaviour through **Average Order Value (AOV)** and **Average Pizza per Order**.
- Provide visualization dashboards for effective decision-making.

Data Source & Description

Dataset: pizza_sales.csv

Key fields:

- **order_id** → Unique identifier for each order
- **pizza_id** → Unique identifier for each pizza
- **pizza_name** → Name of the pizza sold
- **quantity** → Number of pizzas sold per order
- **total_price** → Total revenue for each transaction
- **date, time** → Order timestamp for time-based analysis
- **pizza_category, pizza_size** → Attributes for pizza classification

Tools Used

Python

KPI Analysis

KPI's

```
total_revenue = df['total_price'].sum()
total_pizzas_sold = df['quantity'].sum()
total_orders = df['order_id'].nunique()
avg_order_value = total_revenue/total_orders
avg_pizza_per_order = total_pizzas_sold/total_orders


# Displaying the results
print(f'Total Revenue: {total_revenue:.2f}$ ')
print(f'Total Pizzas Sold: {total_pizzas_sold}')
print(f'Total Orders: {total_orders}')
print(f'Average Order Value: {avg_order_value:.2f}$')
print(f'Average Pizza Per Order: {avg_pizza_per_order:.2f}')
```

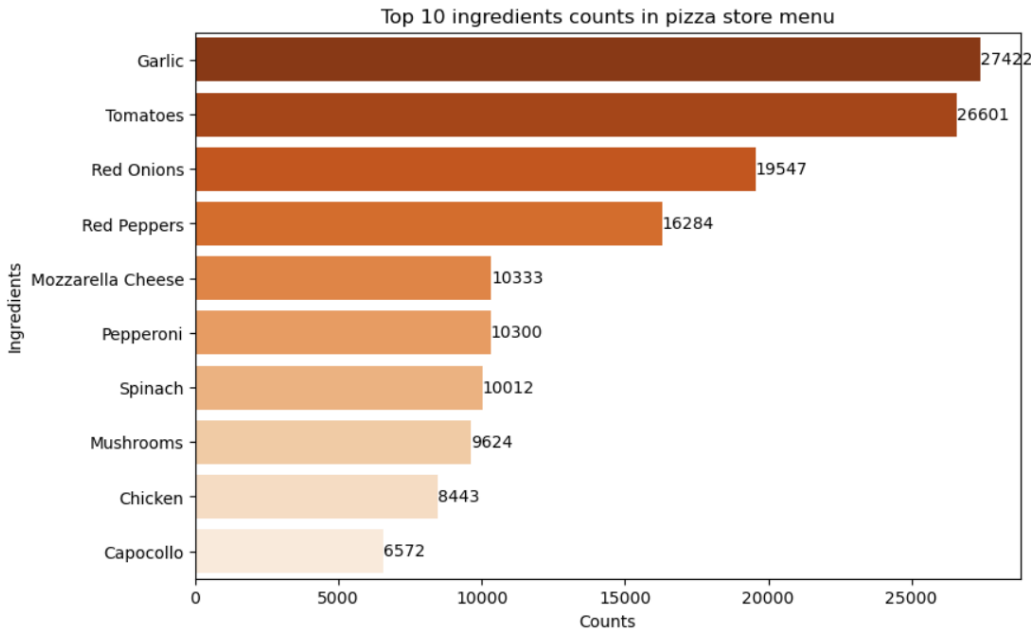
```
Total Revenue: 817860.05$
Total Pizzas Sold: 49574
Total Orders: 21350
Average Order Value: 38.31$
Average Pizza Per Order: 2.32
```

Ingredients Analysis

Top 10 Ingredients

Top_10_ingredients

	pizza_ingredients	counts
0	Garlic	27422
1	Tomatoes	26601
2	Red Onions	19547
3	Red Peppers	16284
4	Mozzarella Cheese	10333
5	Pepperoni	10300
6	Spinach	10012
7	Mushrooms	9624
8	Chicken	8443
9	Capocollo	6572



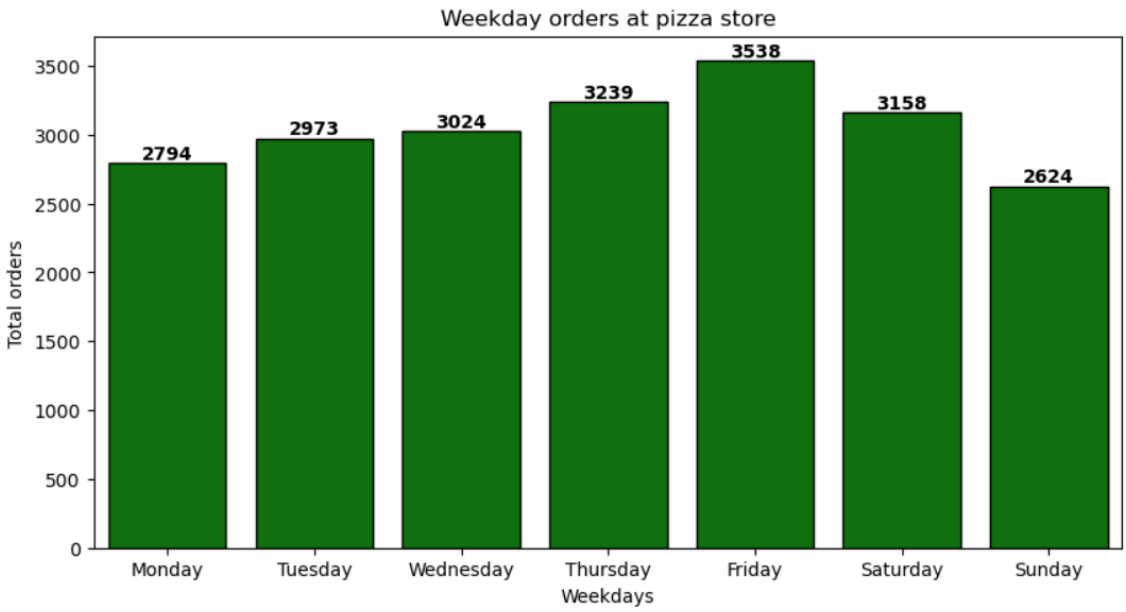
Daily Trend Analysis:

On basis of Total Orders

```
#Grouping order days with total orders
day_wise_orders = df.groupby('order_days')['order_id'].nunique().reset_index()

# Plotting bar plot to visualize the results
plt.figure(figsize=(10,5))
ax=sns.barplot(day_wise_orders,
            x='order_days',
            y='order_id',
            color='green',
            edgecolor='black')
for i in ax.containers:
    ax.bar_label(i,fontweight='bold',label_type='edge')

plt.title('Weekday orders at pizza store')
plt.xlabel('Weekdays')
plt.ylabel('Total orders')
plt.show()
```

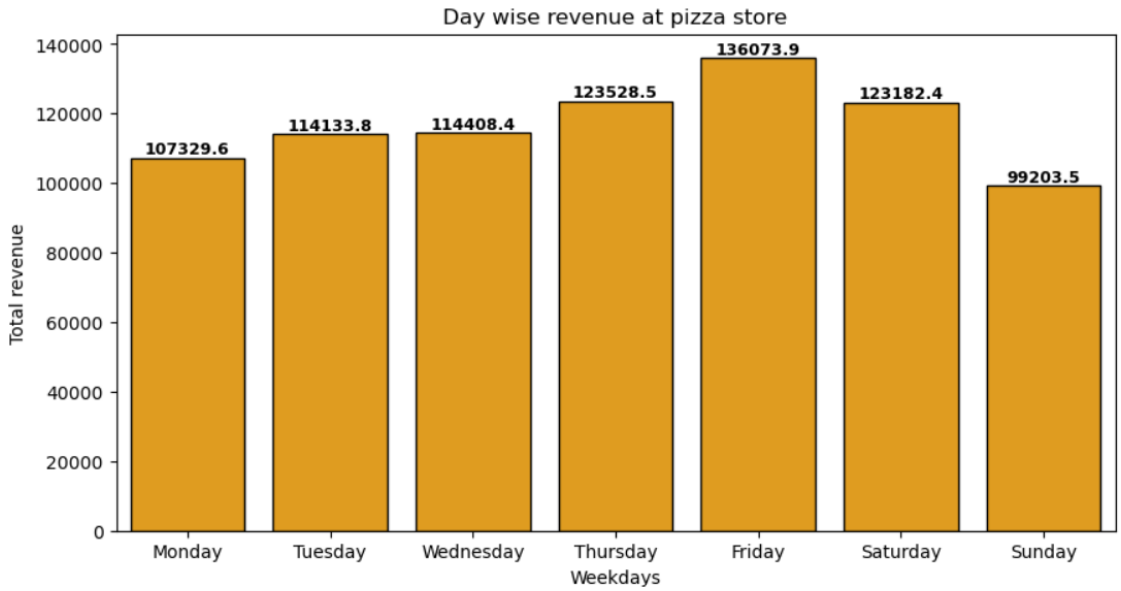


On basis of Total Revenue

```
# Grouping weekdays with total revenue
day_wise_revenue = df.groupby('order_days')['total_price'].sum().reset_index()

#Plotting the bar plot to visualize the result
plt.figure(figsize=(10,5))
ax=sns.barplot(day_wise_revenue,
               x='order_days',
               y='total_price',
               color='orange',
               edgecolor='black',
               errorbar=None)
for i in ax.containers:
    ax.bar_label(i,fontweight='bold',fmt='%.1f',fontsize=9,label_type='edge')

plt.title('Day wise revenue at pizza store')
plt.xlabel('Weekdays')
plt.ylabel('Total revenue')
plt.show()
```



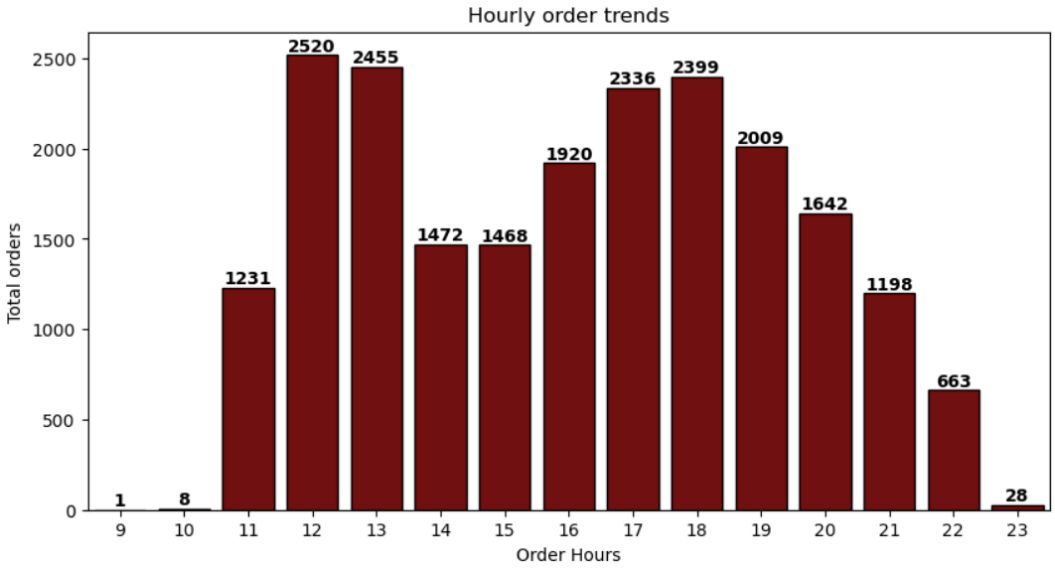
Hourly Trend:

On basis of Total Orders

```
hourly_orders = df.groupby('order_hour')['order_id'].nunique().reset_index()

# Plotting barplot to visualize the results
plt.figure(figsize=(10,5))
ax=sns.barplot(hourly_orders,
               x='order_hour',
               y='order_id',
               color='maroon',
               edgecolor='black',
               errorbar=None)
for i in ax.containers:
    ax.bar_label(i,fontweight='bold',fmt='%.0f',fontsize=10,label_type='edge')

plt.title('Hourly order trends')
plt.xlabel('Order Hours')
plt.ylabel('Total orders')
plt.show()
```

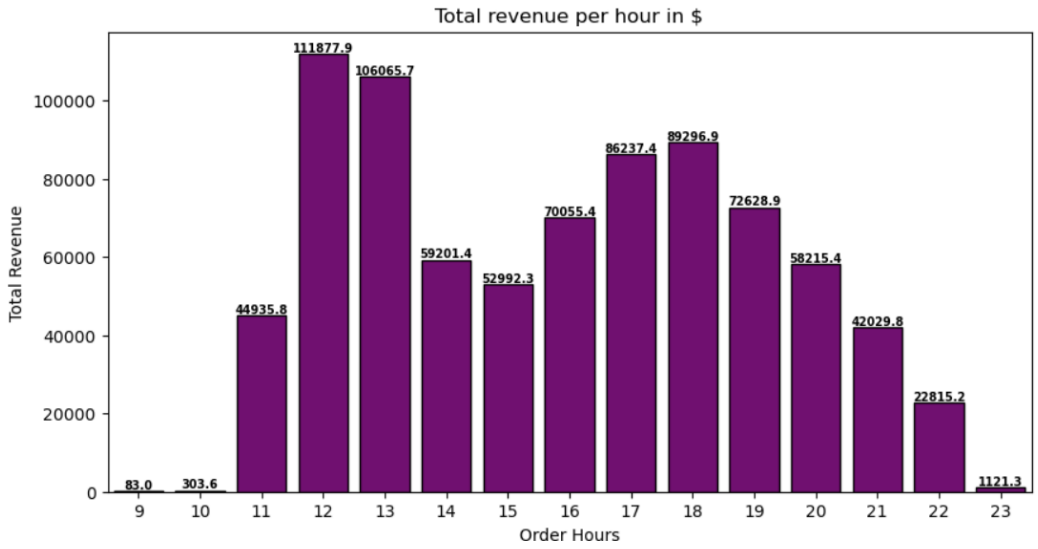


On basis of Total Revenue

```
revenue_per_hour = df.groupby('order_hour')['total_price'].sum().reset_index()

#Plotting the results
plt.figure(figsize=(10,5))
ax=sns.barplot(revenue_per_hour,
            x='order_hour',
            y='total_price',
            color='purple',
            edgecolor='black',
            errorbar=None)
for i in ax.containers:
    ax.bar_label(i,fontweight='bold',fmt='%.1f',fontsize=7,label_type='edge')

plt.title('Total revenue per hour in $')
plt.xlabel('Order Hours')
plt.ylabel('Total Revenue')
plt.show()
```



Monthly Trend

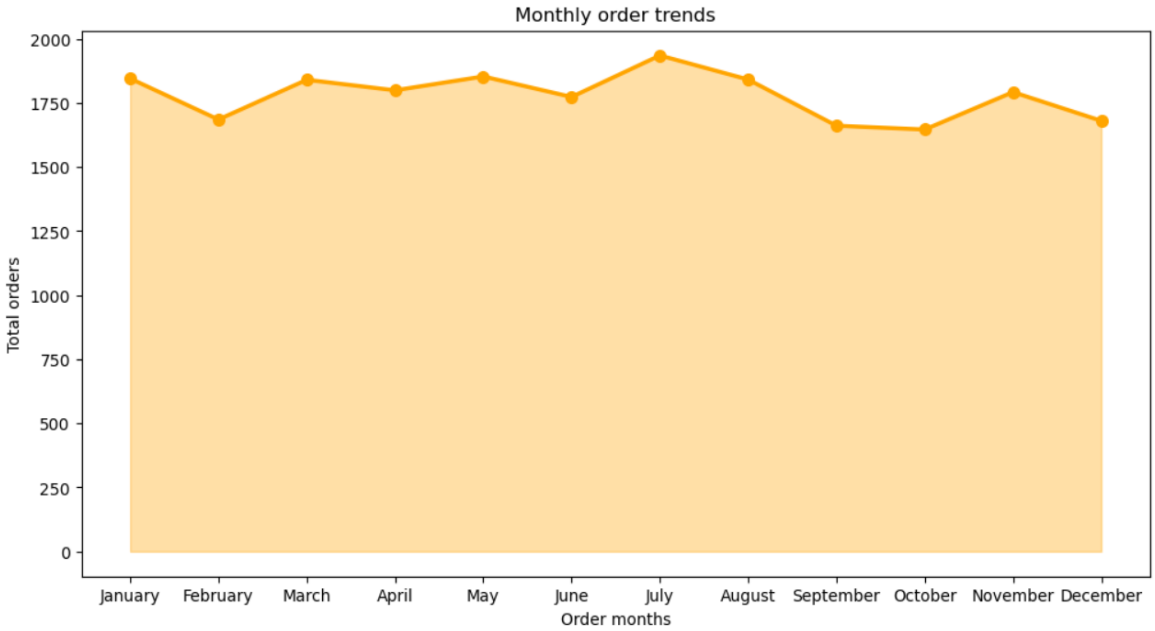
On the basis of Total orders

```
monthly_orders = df.groupby('order_month')['order_id'].nunique().reset_index()

#Plotting line plot to indicate monthly order trends
plt.figure(figsize=(12,6))
plt.plot(
    monthly_orders['order_month'],
    monthly_orders['order_id'],
    color='orange',
    marker='o',
    markersize=7,
    linewidth=2.5,
    label='Monthly Orders')


plt.fill_between(
    monthly_orders['order_month'],
    monthly_orders['order_id'],
    color='orange',
    alpha=0.35)

plt.title('Monthly order trends')
plt.xlabel('Order months')
plt.ylabel('Total orders')
plt.show()
```



On the basis of Total Revenue

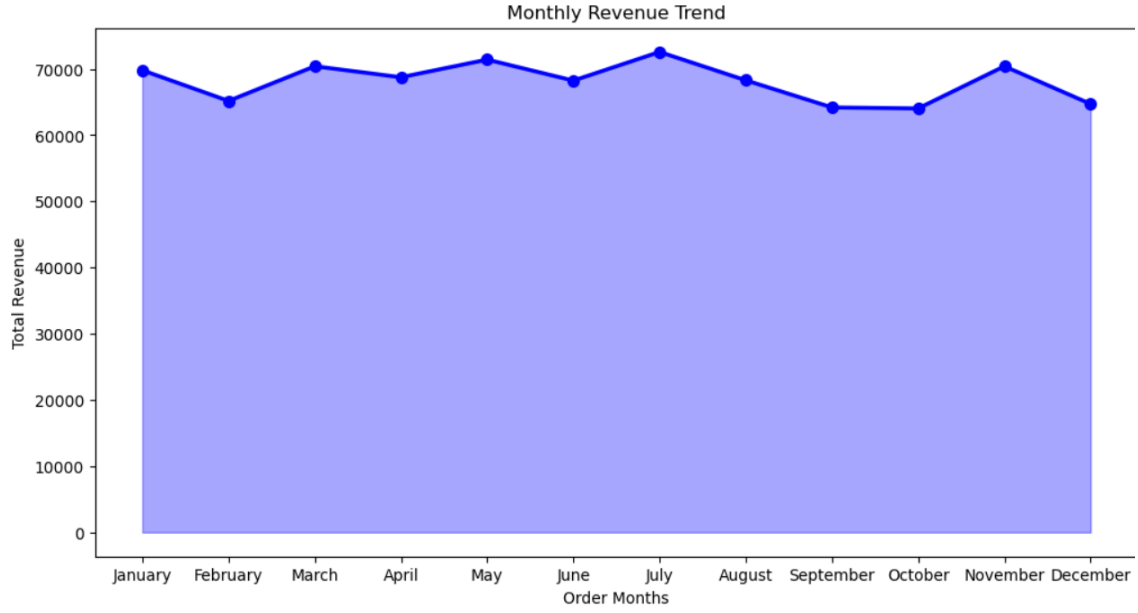
```
month_wise_revenue = df.groupby('order_month')['total_price'].sum().reset_index()

#Plotting a line chart
plt.figure(figsize=(12,6))

plt.plot(
    month_wise_revenue['order_month'],
    month_wise_revenue['total_price'],
    marker='o',
    markersize=7,
    color='blue',
    linewidth=2.5,
    label='Monthly Revenue')

plt.fill_between(
    month_wise_revenue['order_month'],
    month_wise_revenue['total_price'],
    color='blue',
    alpha=0.35)

plt.title('Monthly Revenue Trend')
plt.xlabel('Order Months')
plt.ylabel('Total Revenue')
plt.show()
```



% sales by category

```
pizza_category_sales = df.groupby('pizza_category')['total_price'].sum().reset_index()

pizza_category_sales['Total_revenue'] = pizza_category_sales['total_price'].sum()

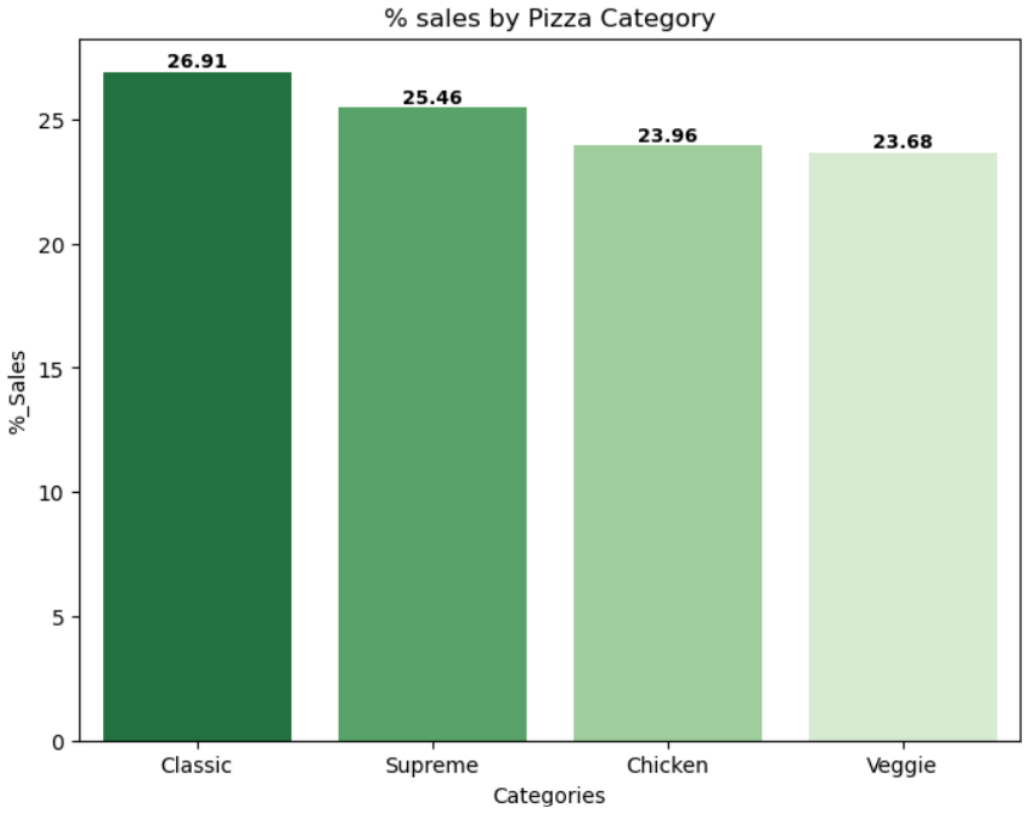
pizza_category_sales['%_sales'] = (pizza_category_sales['total_price']/pizza_category_sales['Total_revenue'])*100

pizza_category_sales['%_sales'] = pizza_category_sales['%_sales'].round(2)

pizza_category_sales.sort_values(by='%_sales',ascending=False,inplace=True)

pizza_category_sales
```

	pizza_category	total_price	Total_revenue	%_sales
1	Classic	220053.10	817860.05	26.91
2	Supreme	208197.00	817860.05	25.46
0	Chicken	195919.50	817860.05	23.96
3	Veggie	193690.45	817860.05	23.68

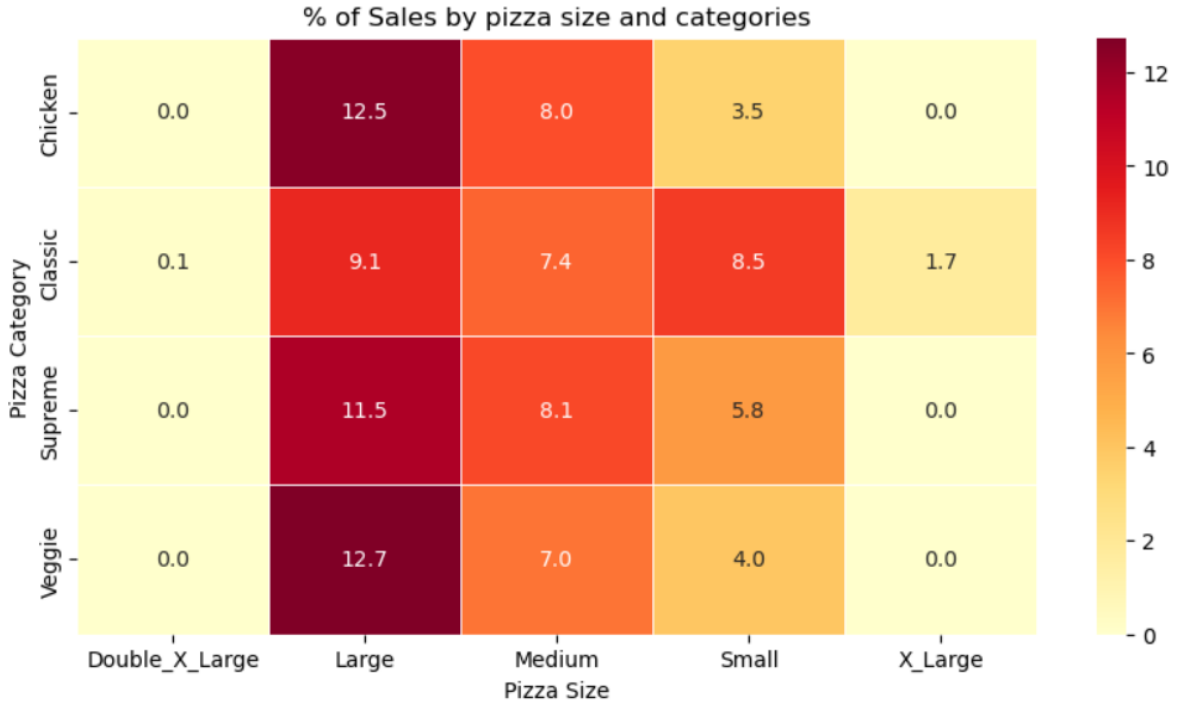


% sales by Pizza size and category

```
sales_pivot=df.pivot_table(
    index='pizza_category',
    columns='pizza_size',
    values='total_price',
    aggfunc='sum',
    fill_value=0 )
```

```
sales_pct= sales_pivot/sales_pivot.sum().sum()*100
```

```
#Plotting a heatmap to display the results
plt.figure(figsize=(10,5))
sns.heatmap(sales_pct,annot=True,fmt='.1f',cmap='YlOrRd',linewidths=0.5)
plt.title('% of Sales by pizza size and categories')
plt.xlabel('Pizza Size')
plt.ylabel('Pizza Category')
plt.show()
```



Total Pizzas sold by pizza categories

```
category_wise_pizza_sold = df.groupby('pizza_category')['quantity'].sum().reset_index()

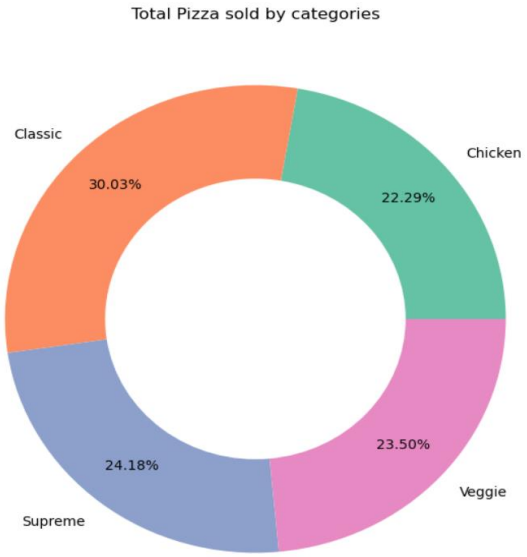
category_wise_pizza_sold['Total_quantity'] = category_wise_pizza_sold['quantity'].sum()

category_wise_pizza_sold['%_quantities'] = (category_wise_pizza_sold['quantity']/category_wise_pizza_sold['Total_quantity'])*100

category_wise_pizza_sold['%_quantitites'] = category_wise_pizza_sold['%_quantities'].round(2)

category_wise_pizza_sold
```

	pizza_category	quantity	Total_quantity	%_quantities	%_quantitites
0	Chicken	11050	49574	22.289910	22.29
1	Classic	14888	49574	30.031872	30.03
2	Supreme	11987	49574	24.180014	24.18
3	Veggie	11649	49574	23.498205	23.50



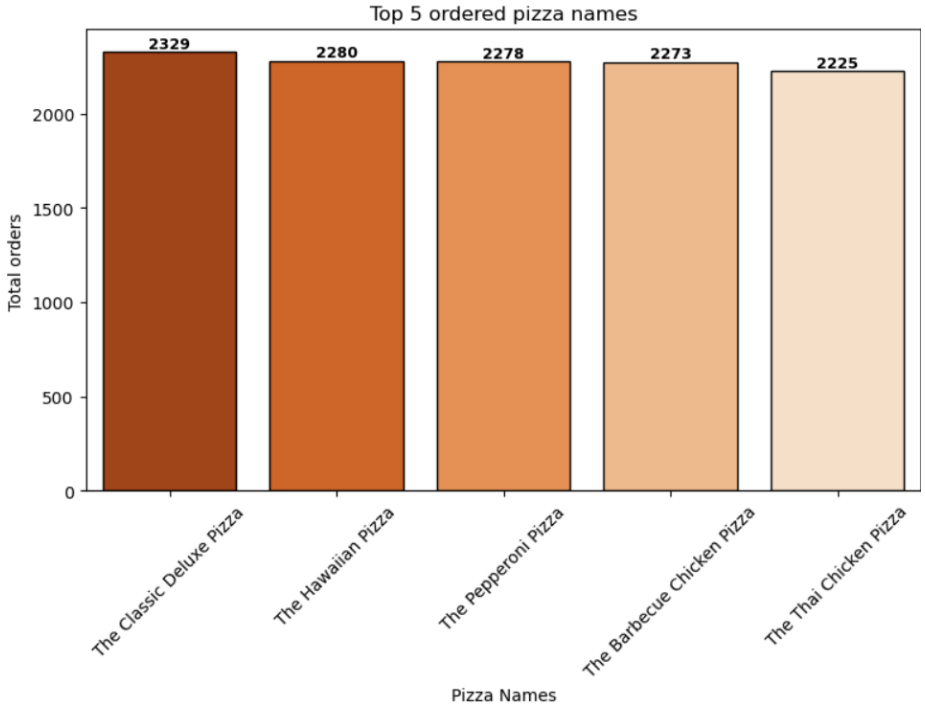
Top 5 best-selling pizza:

On basis of orders and revenue

```
Top_5_ordered_pizza = df.groupby('pizza_name')['order_id'].nunique().sort_values(ascending=False).head().reset_index()

Top_5_ordered_pizza
```

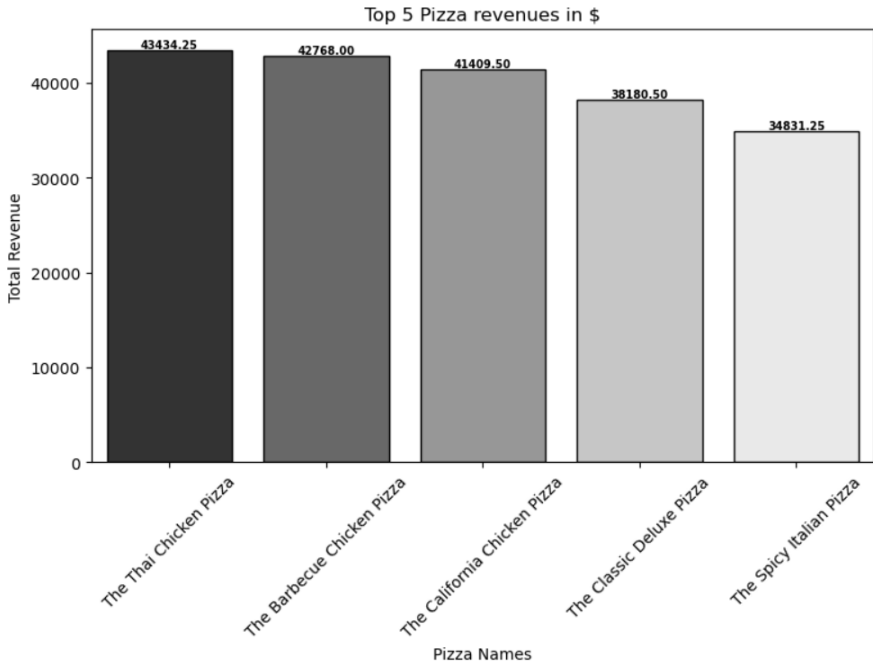
	pizza_name	order_id
0	The Classic Deluxe Pizza	2329
1	The Hawaiian Pizza	2280
2	The Pepperoni Pizza	2278
3	The Barbecue Chicken Pizza	2273
4	The Thai Chicken Pizza	2225




```
Top_5_pizza_revenues = df.groupby('pizza_name')['total_price'].sum().sort_values(ascending=False).head().reset_index()

Top_5_pizza_revenues
```

	pizza_name	total_price
0	The Thai Chicken Pizza	43434.25
1	The Barbecue Chicken Pizza	42768.00
2	The California Chicken Pizza	41409.50
3	The Classic Deluxe Pizza	38180.50
4	The Spicy Italian Pizza	34831.25



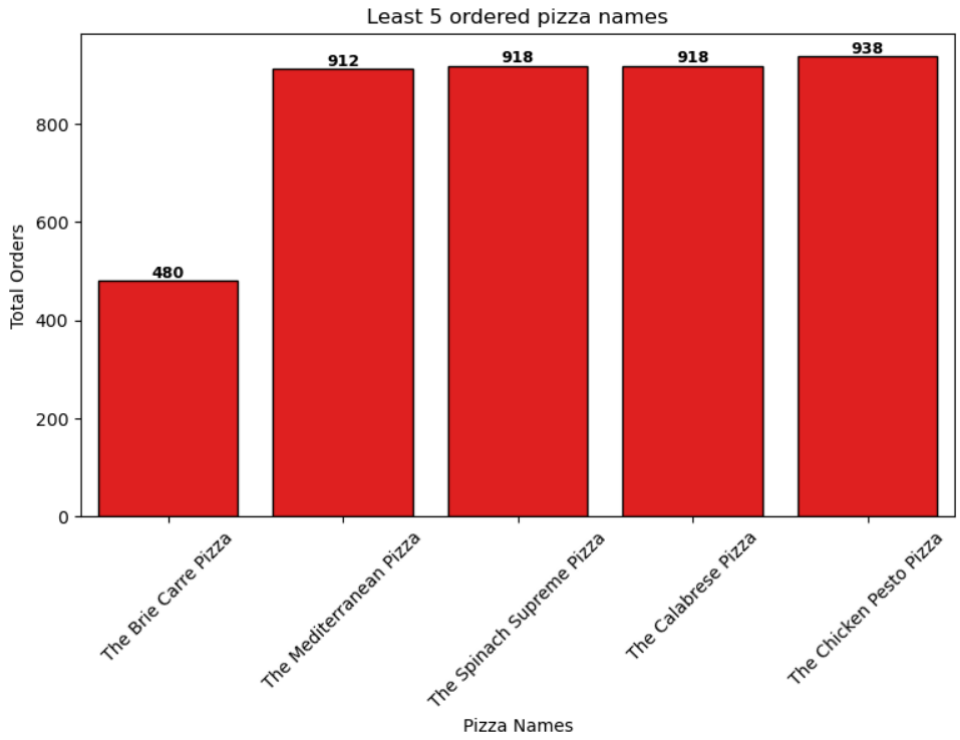
Bottom 5 selling pizza

In terms of orders and revenues

```
least_5_ordered_pizza = df.groupby('pizza_name')['order_id'].nunique().sort_values(ascending=True).head().reset_index()

least_5_ordered_pizza
```

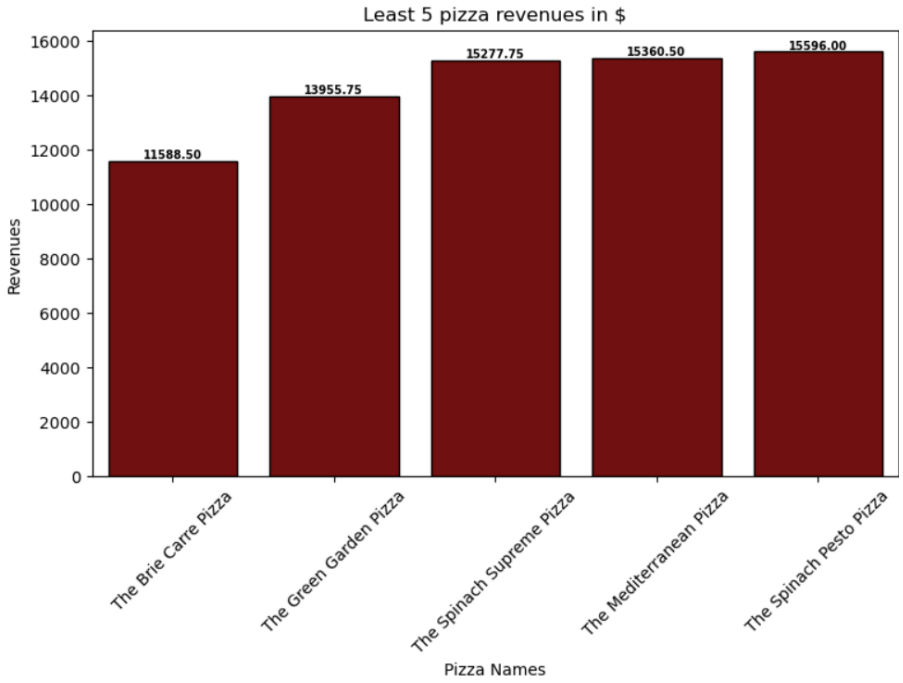
	pizza_name	order_id
0	The Brie Carre Pizza	480
1	The Mediterranean Pizza	912
2	The Spinach Supreme Pizza	918
3	The Calabrese Pizza	918
4	The Chicken Pesto Pizza	938




```
least_5_pizza_revenue = df.groupby('pizza_name')['total_price'].sum().sort_values(ascending=True).head().reset_index()

least_5_pizza_revenue
```

	pizza_name	total_price
0	The Brie Carre Pizza	11588.50
1	The Green Garden Pizza	13955.75
2	The Spinach Supreme Pizza	15277.75
3	The Mediterranean Pizza	15360.50
4	The Spinach Pesto Pizza	15596.00



Key Findings:

- The total revenue generated from the pizza store is 817860.05 $.
- Total 49574 pizzas were sold in 2015 from January to December.
- Veggie and Chicken category of large size have been the highest revenue generating pizzas amongst all the categories and sizes.
- **Top performer pizzas-** The classic deluxe pizza**,** The barbeque chicken pizza, The Thai chicken pizza.
- **Bottom performer pizzas**-The brie Carre pizza, The spinach supreme pizza, The mediterranean pizza.
- The average order value of the pizza is 38.31$ and the average pizza ordered per order is 2.32.
- **Sales and order trends on weekdays basis** – Thursday, Friday and Saturday have been the most profitable days for the entire year in 2015 as it has witnessed maximum number of orders and revenue.
- **Sales and order trends on hourly basis**- It has been witnessed that in a day the afternoon time 12-1 PM and the evening time 4-6 PM have been the busiest hours for the pizza store as chart indicates spike in the orders in those hours.
- **Sales and order trends on monthly basis**- The performance of pizza store has almost been consistent for each of the month in 2015 however, There is just a slight increase in the orders and revenue in the month of July.